

Sprint 025: Resolution Tier (0–15) Boundary Check & Thermodynamic Invariants in Hierarchical Spatial Discretization

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Sprint 025 — Repository: <https://github.com/pascalranoroarijaona/WebOfLife>

Abstract

In computational systems ecology and planetary-scale biosphere simulation, spatial discretization dictates the granularity of biogeochemical cycling, energy flux distribution, and trophic network interactions. Sprint 025 establishes the formal resolution tier boundary check function for Uber’s H3 hierarchical hexagonal spatial index within the *Web of Life* engine. Restricting spatial resolution tiers strictly to the integer domain $r \in [0, 15]$ prevents topological corruption and maintains rigorous thermodynamic compliance. This paper outlines the theoretical framework governing spatial refinement, mass conservation (First Law of Thermodynamics), and solar energy flux allocation (Second Law of Thermodynamics) under monadic state transitions.

1 Introduction & Theoretical Context

The *Web of Life* simulation engine models planetary ecosystems by coupling biogeochemical cycles with discrete spatial topologies. Spatial partitioning is implemented via Uber’s H3 hierarchical hexagonal grid system, which partitions the Earth’s surface across 16 discrete resolution tiers ($r \in [0, 15]$).

Within this architecture, spatial tiers are not merely geometric coordinates; they dictate ecological carrying capacities and metabolic rates. Refining spatial resolution from tier r to $r+1$ increases spatial precision by a scaling factor tied to H3’s average hexagonal area contraction ratio:

$$\bar{A}_{r+1} \approx \frac{\bar{A}_r}{7} \quad (\text{exact area scaling factor} \approx 7.3735 \text{ depending on tier}) \quad (1)$$

To prevent arbitrary dimensional scaling that violates physical conservation laws, Sprint 025 implements strict runtime validation bounds (`isValidH3Resolution` and `assertValidH3Resolution`) integrated directly into the `SpatialMonad` execution pipeline.

2 Mass & Energy Conservation Equations

2.1 Spatial Partitioning Mass Conservation (First Law)

When a parent hexagonal stock at resolution r containing mass stocks M_{carbon} , M_{water} , and M_{minerals} is refined into child cells at resolution $r+1$, total elemental mass must be conserved exactly across all k sub-cells:

$$\sum_{i=1}^k M_{\text{element},i}^{(r+1)} = M_{\text{element, parent}}^{(r)} \quad (2)$$

2.2 Thermodynamic Energy Flux Allocation (Second Law)

Energy inputs into spatial nodes are restricted strictly to solar radiation fluxes (E_{solar}) modulated by geographic latitude and resolution area:

$$\Delta E_{\text{in}} = \Phi_{\text{solar}} \cdot A(r) \cdot \cos(\theta_{\text{lat}}) \quad (3)$$

Refinement ($r \rightarrow r+1$) scales down the surface area $A(r)$, proportionally reducing the localized solar energy flux per node unless metabolic networks aggregate workloads.

3 Implementation & Architecture

The boundary validation logic is implemented in `src/spatial/h3_grid.ts` and governed by types defined in `src/spatial/h3_types.ts`:

Listing 1: H3 Resolution Boundary Validation in TypeScript

```
export type H3Resolution =  
  | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7  
  | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15;  
  
export function isValidH3Resolution(resolution: number): resolution is  
  H3Resolution {  
  return Number.isInteger(resolution) && resolution >= 0 && resolution  
    <= 15;  
}  
  
export function assertValidH3Resolution(resolution: number): asserts  
  resolution is H3Resolution {  
  if (!isValidH3Resolution(resolution)) {  
    throw new RangeError(`Invalid H3 resolution tier: ${resolution}.  
      Resolution must be an integer between 0 and 15.`);  
  }  
}
```

This validation is enforced within the `SpatialMonad` (`src/monads/spatial_monad.ts`), ensuring that all spatial transformations, neighborhood lookups, and compaction routines operate exclusively on valid tier boundaries.

4 Conclusion & Future Work

Sprint 025 successfully secures the foundational spatial tier boundaries of the *Web of Life* simulation engine. By binding geometric indexing directly to thermodynamic conservation laws, the architecture ensures physical realism in planetary-scale ecological simulations. Future sprints will expand on multi-tier trophic energy redistribution and dynamic load balancing across hexagonal partitions.

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